

Vijaya Sai Kalivarapu

+91 9948332933 | vijaysaikalivarapu@gmail.com

<https://github.com/2300033794> | <https://www.linkedin.com/in/vijay-sai-kalivarapu>

SUMMARY

B.Tech Computer Science & Engineering student at KL University (CGPA: 8.89/10) with hands-on experience in full-stack development, AI/ML systems, and DevOps pipelines. Proficient in **Java**, **Python**, **JavaScript**, with practical project experience using **Spring Boot**, **React.js**, **FastAPI**, **MySQL**, **Docker**, and **Kubernetes**. Built and deployed real-time deep learning systems and scalable REST API backends.

EDUCATION

2023 – 2027 **B.Tech in Computer Science and Engineering at Koneru Lakshmaiah Education Foundation (K.L. University), Guntur** (CGPA: 8.89/10)
Specializing in Deep Learning and Full-Stack Development. Proficient in Python, Java, and C, with certified expertise from NVIDIA and HackerRank. Experienced in team collaboration and developing software solutions; eager to apply algorithmic knowledge to real-world problems.

EXPERIENCE

Technical Team Member — Kognitiv Club, KL University *Aug 2024 – Present*

- Analysed and preprocessed datasets of **10,000+ records** to support AI/ML model training, improving data quality through feature engineering and outlier removal to enhance prediction accuracy.
- Designed and deployed **3+ responsive web components** using React.js integrated with REST API backends, delivering features across iterative development cycles and ensuring cross-browser compatibility.
- Collaborated within a **technical team** on agile software development, participating in code reviews, debugging sessions, and structured knowledge-sharing workshops to drive continuous improvement.

PROJECTS

Panoptic Segmentation & Scene Understanding System <https://github.com/2300033794/panoptic-segmentation>

- Built a real-time video analytics system using **YOLOv8** deep learning model, achieving **90%+ object detection accuracy** across multi-class video streams with object detection, segmentation, and scene analysis.
- Developed a **scalable FastAPI** backend with asynchronous request handling, integrated an **LLM-based** object-tracking module, and exposed RESTful endpoints consumed by a **React.js** dashboard.
- Implemented panoptic segmentation and scene-understanding pipelines enabling semantic and instance-level labelling, reducing manual annotation effort.

Research Journal Management System <https://github.com/2300033794/Research-journal-management>

- Architected and delivered a **full-stack web application** using **Spring Boot** (backend), **React.js** (frontend), and **MySQL**, featuring structured manuscript workflows and **role-based access control (RBAC)**.
- Designed and implemented **RESTful APIs** enabling structured submission, peer-review, and status-tracking workflows, improving process efficiency.
- Optimised database schema and queries, improving system scalability and reducing redundancy across interconnected data entities.

TECHNICAL SKILLS

Languages	Java, Python, C, C++, JavaScript, SQL
Frameworks	Spring Boot, React.js, Node.js, FastAPI, REST APIs
AI / ML	YOLOv8, Deep Learning, LLM Integration, Data Analysis, Neural Networks
Databases	MySQL, DBMS, Database Schema Design, Query Optimisation
DevOps	Docker, Kubernetes, Ansible, Git, GitHub, CI/CD, Linux
Core CS	Data Structures & Algorithms, OOP, Operating Systems, Computer Networks

CERTIFICATIONS

Fundamentals of Deep Learning	NVIDIA — Training techniques, model optimisation, neural network architectures
Python Basics	HackerRank — Core Python, algorithmic problem-solving
Python for Data Science & AI	IBM — Data analysis, visualisation (Pandas, NumPy), AI fundamentals